



**Department of Electrical and Computer Engineering
North South University**

Senior Design Project

AI meets Law: Legal vs Illegal Activity Detection and Justification

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LETTER OF TRANSMITTAL

December, 2024

To

Dr. Mohammad Abdul Matin
Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Capstone Project Report on “AI meets Law: Legal vs Illegal Activity Detection and Justification”

Dear Sir,

With due respect, we would like to submit our **Capstone Project Report** on “**AI meets Law: Legal vs Illegal Activity Detection and Justification**” as a part of our BSc program. This project has been incredibly valuable to us as it provided an opportunity to explore the intersection of Artificial Intelligence and legal analysis, applying advanced Retrieval-Augmented Generation (RAG) models to the Penal Code of 1860. By working on this project, we gained hands-on experience in legal text classification and explanation generation, which are critical in making legal documents more accessible and understandable. We tried to the maximum competence to meet all the dimensions required from this report.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

.....

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APPROVAL

Md. Abul Kashem Junaid (ID # 2021867642), Md. Hasibul Alam Chowdhury (ID # 2012692042), Muhammad Imam Hassan (ID # 2013362042) and Md. Abdur Rauf (ID # 1821033042) from Electrical and Computer Engineering Department of North South University, have worked on the Senior Design Project titled “AI meets Law: Legal vs Illegal Activity Detection and Justification” under the supervision of Dr. Mohammad Ashrafuzzaman Khan partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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ABSTRACT

AI meets Law: Legal vs Illegal Activity Detection and Justification

This study proposes a framework to automate legal reasoning and decision making on the Penal Code of 1860. Our proposed system generates using RAG style, combining the precise retrieval capabilities of Gemma2b and the generative capabilities of Llama3.2 for generative reasoning. It classifies scenarios provided by users as legal or illegal and justifies the reasoning behind it according to certain relevant legal sections. For legal actions, the system provides rationales. If illegal, it lays out the violations and how offenders are punished. Legal texts are chunked into context-preserving segments, encoded into dense vector embeddings, and made possible for efficient retrieval using SentenceTransformers, ChromaDB and similar tools. User queries are embedded and matched against the most relevant legal provisions to feed them into the Llama3.2. This design connects static searching of legal documents with dynamic searching for scenario-based analysis and provides specific, actionable insights for the user. The system's capabilities were tested with scenarios such as self-defense and intoxication and demonstrated its potential to help legally trained professionals with case assessment and to make legal knowledge available to non-experts. Increased coverage of case laws, amendments, and legal frameworks from various jurisdictions is an avenue of future work. Features like query intent detection, adaptive learning, and multilingual support will yield improved accuracy and usability. This project demonstrates how AI can transform legal knowledge and research to make it accessible and useful to a variety of people.

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Chapter 1 Introduction

1.1 Background and Motivation

The legal field is perceived as complex and hard to navigate around, something closely related to the abstract language and terminology as well as the abstract reasonings that characterize legal texts. This complexity is a hurdle for non-experts as well as for people who are attempting to self-represent in lawsuits or need quick legal advice but do not have formal access to legal representation. Legal texts (statutes, regulations, case law, precedents) are difficult to interpret and require specialized knowledge for proper application. These complexities become even starker in countries like Bangladesh, whose penal law is essentially built on the foundation set in 1860. For a work of legislation, the Penal Code is an expensive and historical artifact, full of jargon and vague provisions that require careful parsing, particularly applied to real-time legal circumstances.

This is especially difficult, however, in the case of the Penal Code of 1860. The Code defines what conduct is considered a criminal offense and the penalties associated with it, and is extensive, but it is still not easy for the average citizen to understand. It relies on technicalities of criminal liability, self-defense, intoxication, and other topics that turn people into characters in a sitcom, not a courtroom. Without due process of law with limited knowledge, the law can punish some wrongfully have been in the wrong place just like having a opinion and justly accepting false accusations wrong is sometimes all they would have said. In jurisdictions with limited access to legal practitioners, this can be further compounded as a result of individuals finding it challenging to source the correct advice or know where to start in their search for answers.

This creates a huge divide between the public knowledge of the law and the legal awareness needed to act appropriately under numerous circumstances. Legal professionals, despite their training, are frequently overburdened and face a workload that keeps increasing; the number of legal texts that must be processed and studied is nearly infinite, and it slows down the pace of justice. Finding relevant statutes, understanding their nuances, and explaining or applying them to clients is an error-prone, highly time-consuming process. For the general public, this legal distance describes a detachment between the law and the prevention of legal violation which in turn treats the law as a far-fetched concept that everyone cannot access thus creating a form of injustice.

This shows there is a real need for tools that can simplify and democratize legal knowledge for non-experts. But legal research, the traditional art of scouring hundreds, if not thousands, of pages of legal texts by hand, is both tedious and time-consuming. The enormous amount of material the lawyers need to read and understand during the entire process makes every single of them cumbersome and slavery even for experienced ones. This dogged research process is especially time-consuming when it comes to thorny legal questions, like whether a person has a right to use deadly force when they think their life is in danger, or whether someone under the influence of drugs or alcohol can be held criminally liable for their actions. All this legal doctrine requires contextually individualized assessments of the law that cannot be easily mined without intensive exploration of the law.

Every bifurcation of law with diverse case studies makes a lag in available legal resources for the public and legal professionals; it's a need of an hour for an innovation that can make it easy to research and analyze the needs in very less time. Our aim is to overcome these barriers with this project where we aim at creating a tool-based solution to help automate the process of legal analysis using RAG models and LLMs, making it more accessible for anyone whether they are lawyers or not to reason about the law. - To provide contextually relevant legal responses immediately depending on the users' questions related to wide variate of legal scenarios under which the penal code of 1860 will be a center,

This is where AI technologies like RAG models become key players. RAG models are fundamentally different from the conventional information retrieval methods used in massive datasets because the retrieval process is usually based on keyword matching, and then the information is generated and returning the searched episode in the relevant detail but unable to develop contextually relevant information. The integration of retrieval and generation capabilities allows RAG to excel in multi-faceted, context-rich tasks such as legal analysis, which requires a nuanced understanding of specific case facts and the relevant legal statutes.

For example, in this project, when I ask the system whether carrying a firearm for self-defense is allowed under the Penal Code of 1860, it retrieves the relevant legal provisions but more importantly, it generates a clear explanation of how these provisions apply to my specific scenario.

This can save legal professionals tons of time spent on research, as the automation of this process will free up more time spent on strategic decision-making and representation of clients. Moreover, the system's capacity to generate contextualized responses also renders it an invaluable asset for teaching lawyers and lawyers-in-training, as it can assist them in swiftly grasping the practical consequences of legal concepts.

The motivation behind this research is several-fold. The increasing demand for tools that enable easy access to legal knowledge for the public, especially in areas where legal services are scarce. Such tools can help fill this gap and provide individuals with immediate, actionable legal information, in a country such as Bangladesh, where legal professionals are often scarce and really less in rural areas. This framework can provide people with detailed, real-world explanations about their rights and responsibilities, helping them make informed decisions.

Furthermore, this project is inspired by the ability of AI to revolutionize the way we perform legal research. The traditional method of conducting legal research is both time and labor-intensive. Lawyers spend hours combing through hundreds of thousands of pages of legal language to find relevant statutes or case law. Once provisions that are still relevant have been located, they must still be interpreted and applied to the particular facts of a c... This project will significantly streamline legal research by automating both the retrieval of relevant legal information and the generation of legal responses. The efficiency gains can allow legal professionals to dedicate time and attention to the higher-level elements of their work, like case strategy, negotiation, and counseling.

Also, this could help improve access to the law for those who cannot afford an lawyer or cannot access legal profession in some areas. It will help people quickly determine the legality of their conduct, whether carrying a firearm or using calculative resort of force in self-defense is permissible under the Penal Code of 1860. This can lessen dependency on costly legal services and encourage people to have more confidence in knowing their legal rights and duties. Another way that this technology helps to support users is by making legal information more accessible to the masses, i.e. someone without a formal legal background can also be able to grasp the concept of law in the context of their case.

In building this knowledge graph, we are motivated by the need to enable the citizens of Bangladesh who may face geographic or financial barriers to accessing legal resources. The 1860 Penal Code encompasses a variety of criminal offences, ranging from theft to murder, but its application can be esoteric to those who are not well-versed with legal jargon or reasoning. Legal ideas, such as self-defense, intoxication, etc., concerning liability for crimes, are not always easy to understand. This is a classic area where AI, especially through RAG models, can assist in demystifying these dense provisions, crafting explicit and actionable legal recourses specific to a particular context.

Also, the project is motivated by future enhancements of the system. Further, we can scale the dataset as the system is developed — to include more legal provisions, case law and even snapshots of amendments to the Penal Code of 1860. The system may also be improved to the more complex legal cases, including multi-jurisdictional matters, and integrate multi-lingual allowances to help serve a wider population. Such feedback loops would also be able to allow the general public to suggest improvements to the model, helping increase the accuracy and context of the classification system and ensuring the responses generated are as accurate and helpful as possible.

This solution aims to combat the critical demand for legal devices that are efficient, attainable, and precise. It will build a system that leverages RAG models and LLMs to reshape the process of legal research, making legal knowledge available to legal professionals and the general public., presenting an work set to the marginalization of legal analyses, and opening the door to the development of a universal understanding of the public about the law, which is likely to reduce dependence on professional lawyers in Bangladesh. It is such a tool to revolutionize the legal research supplementary system and empower people with knowledge and bridge the gap between the intricacies of the law and the people who must live with it.

1.2 Purpose and Goal of the Project

This project is primarily concerned with the creation of a novel legal analytic system that leverages Retrieval-Augmented Generation (RAG) models and other types of Large Language Models (LLMs) to allow for automation and enhancement of the understanding and application of the Penal Code of 1860. This legal code is a primary text in Bangladesh, affecting all aspects of criminal law, and its complexities and dense legal language can make it difficult to understand for

those without legal training. Project seeks to close this gap by using AI to bottle down legal knowledge into simplified forms, while also enabling you to implement them.

Legal Research and Interpretation Automation: One of the core products of this project is the automation of legal research and interpretation domains via RAG models. Lawyers, judges and law students spend considerable effort sifting through previous legal provisions, precedents and case law manually. The process can be time-consuming and error prone. These models, also referred to as RAG models, will automate the retrieval of relevant sections from the Penal Code of 1860 & produce crisp, context-oriented explanations as per user query. Such as accelerating legal research, saving a considerable amount of time and effort and improving accuracy.

Coherent and Appropriate Descriptions: Conventional legal research could be done through keyword-based search or rule-based matching, however, the new generation approach will build on advanced AI models to generate human-readable context-specific explanations of legal provisions. Example: If a user asks about what the law says for a specific case, referring to whether a soldier fired into a mob without verifying that mob does indeed represent a threat, the system not just fetches sections of the Penal Code, but details why that soldier cannot claim immunity under the law. Such capability to yield in-depth textual explanations that are context-relevant is a first-of-its-kind contribution in legal search and research, because it transforms the complexity of the legal text into more digestible information for all, including lawyers (also known as ‘legal experts’) and non-expert users.

A major objective of this project is the democratization of legal knowledge. In places such as Bangladesh, where access to legal professionals may be scarce or expensive, this system will assist as a means of self-directed legal learning. This AI Example will help people think intelligently about the law and provide them with the ability to make the right decisions without having to hire Task Forces. This is valuable for those who are unable to pay for the services of lawyers or for those who reside in a jurisdiction with little legal coverage.

Navigating their interplays between retrieval and generation for legal domains: What makes this project unique is the use of Retrieval-Augmented Generation (RAG) models which merge the strength of information retrieval with text generation. RAG models are highly suitable for the legal domain, as they enable the system to retrieve relevant legal materials while also generating fluent,

detailed answers for explaining how legislation relates to a particular case. This dual capability is critical for complex legal analysis, as it allows the system to deal with both the breadth (finding relevant texts) and depth (producing context-aware justifications) of the analysis.

Improving Legal Education and Raising Public Awareness: A further key contribution of this project is its alignment with supporting legal education and public awareness. Data is trained until October 2023. The system could generate clear explanations to help users identify how laws like the Penal Code of 1860 apply to certain scenarios, such as criminal liability, self-defense, or intoxication and extreme provocation. This, as it would enable students and trainees to see how the law is implemented and used, beyond theory — would be a contribution that much improves the quality of legal education.

Scalability and Generalization: Although this project is limited to the Penal Code of 1860, its ecosystem is future-proofed and extensible to other legal domains and codes. Utilizing RAG models and LLMs, we could extend the system to cover a multitude of legal databases like case law, changes in local laws, and judicial decisions, offering a more comprehensive address of legal inquiries. The scalability of this system not only aids in understanding the Penal Code of 1860, but also in moving into other laws, as they are developed.

Making Legal Workload for Less Professionals: This project can also be valuable in providing assistive features, which can help the lawyers reduce their workload. In cases with voluminous legal documents at play, the search for applicable legal texts and interpretative work on dense legal wording can be extremely time-consuming. The AI-based technology will save time and effort for legal professionals, as they will no longer have to spend hours on routine research as the system will retrieve and explain legal texts. That efficiency elevating can help transform the law practice, particularly in dynamic law.

Additionally, the focus of the project on the 1860 Penal Code as a significant piece of legal framing in the South Asian context allows for a novel setting to both evaluate the performance of RAG models in the domain of law, as well as exploring their potential. The technical nature and historical complexity of the Penal Code are a challenge for both practitioners and the public, making it an ideal case for exploring the promise of AI to lower the barriers to legal analysis.

We believe that this project can contribute significantly to the landscape of legal technology, by creating an AI-based system that augments legal insights through more effective access to legal knowledge and assists legal professionals in their work. The project will leverage Retrieval-Augmented Generation (RAG) models and other LLMs to automate and facilitate the process of parsing, interpreting, and summarizing the Penal Code of 1860, ensuring accessibility for a broad audience of users. The novelty of this project lies in its high-tech use of state-of-the-art AI technologies to resolve legal research and interpretation issues and the fact that this is expected to transform access, comprehension and application of legal information.

1.3 Organization of the Report

- Chapter 1 represents Introduction.
- Chapter 2 represents Research Literature Review.
- Chapter 3 represents Methodology.
- Chapter 4 represents Investigation/Experiment, Result, Analysis and Discussion.
- Chapter 5 represents Impacts of the project on both Social and Environmental.
- Chapter 6 represents Project Planning and Budget.
- Chapter 7 represents Complex Engineering Problems and Activities
- Chapter 8 represents Conclusion

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

While some of the existing studies about legal text classification, as well as AI for legal services, deliver promising results, they also have some weaknesses that our project addresses directly. Generative Prompting for Legal Text Classification: Zero-, Few- and Chain-of-Thought Prompting with the OPT Family of Language Models Improves LLM performance on legal datasets including the European Court of Human Rights (ECHR) and the Supreme Court of the United States (SCOTUS) Such an approach however is restricted to datasets in English, therefore limiting its potential for deployment in non-English-speaking legal systems. Furthermore, though the study above explored the performance through context-rich prompting, it does not underlie the importance of contextual explanations, or outputs that are relevant concerning the context which evidently becomes more important in application to nuanced laws such as for instance, the Penal Code of 1860 in Bangladesh. Similar to this study, our project utilizes a Retrieval-Augmented Generation (RAG) model; however, instead of outputting relevant modeled legal provisions that are recreated in the Penal Code, these need to be generated on a contextual basis, which explains how those provisions apply to specific cases. And bear in mind that this capacity for producing legal justification is an actually superb point in regards to guaranteeing there will be openness that all those people interested in that do comprehend what the rationale of that decision is not simply an option to that area of issues.

Taking the example of the text classification study in the French legal domain, this work uses machine learning methods to predict the French Supreme Court ruling and classify which law area a case belongs to. While this approach has been effective for legal document categorization, it lacks a focus on how legal provisions are invoked. The scientific work is primarily and obviously concentrated towards document classification where there is and no exploration per se legal reasoning complexities that one needs to comprehend to read the laws for given jurisdiction i.e. the Penal Code of 1860. In contrast, our project fills this gap by developing a system that creates both the legal classification but also a human-readable explanation of how they arrived at that classification with respect to that law. Moreover, the study uses a large corpus of French court rulings, but there is no account of how jurisdiction-specific legal systems can affect the quality of

text, which is an important factor in our project that tries to simplify and explain Bangladesh's Penal Code.

Challenges related to knowledge modeling, legal reasoning, and interpretability are also brought up by the wider domain of Legal AI. Pre-trained language models (PLMs), e.g. BERT, have excelled in various tasks in the legal domain including legal QA and judgment prediction. But a major shortcoming of this research is its blind eye to its own interpretability. While PLMs can read and comprehend substantial amounts of legal text, they normally do not provide easily understandable, explainable reasoning as to the legal conclusions they draw. It is an important question — the mark above all in the instance of the law must those caught up in any given decision be able to clearly identify is the reason for a decision. In our project, we address this limitation by implementing a RAG-based system that retrieves the relevant legal texts and coherently and in context explain them in the context of Portuguese law. Such explanations are crucial for informing users about the reasons underlying legal decisions made -- in complex cases, where other approaches project their limitations, which is usually the case in (complex) legal decision making, explaining the reasoning that led to the decision."

Most recently, a work on large-scale legal text categorization based on transformer models (e.g. BERT and RoBERTa), but the study only addresses multi-label classification brokering with European datasets like JRC-Acquis and EURLEX 57K which is concerned about a tom of documents and in detail, it's only a notch with regard to machine learning, merely the classification, without evidence or explanation about the classifiers in the autopilot. Although these models are widely used for classification on documents, they do not take into consideration the context of application of a specific provision of law — something that is even more important in case of laws like the penal code of 1860. Our project fills this gap, as it combines faithful retrieval of both relevant statutory provisions and facts applicable to the given scenario with a generation process to obtain clear and chain of thought-driven answers to concrete legal questions. Moreover, the European datasets employed in this study while wide-ranging, are not jurisdiction-specific data useful for analysis and understanding of the Bangladesh Penal Code. We adopt a non-English legal system to permit the artificial intelligence cascade to a more granular legal context and a consideration of how necessary it is for legal professionals to comprehend the rationale behind decisions

So, while most work today in Legal AI is useful, it is also largely failing when it comes to adjacent contextualization, jurisdictional embedding, and interpretability of the legal reasoning employed. Unlike past works, our project targets the Penal Code of Bangladesh of 1860 and uses nearest neighbor-based RAG models to produce both legal classes and natural language reasoning. Apart from making legal decisions more accessible and transparent, this also simplifies what tends to be a complex process of interpreting legal provisions and applying them to a specific case and speeds up legal research, for both practitioners and the layperson.

The literature on legal text classification and the utilization of artificial intelligence (AI) in legal services illustrate promising returned results. However, there are numerous limitations in the extant research that our project directly addresses. Generative Prompting for Classifying Legal Text: Zero-, Few- and Chain-of-Thought Prompting with the OPT Family of Language Models Improves LLM performance on END-NLP5854D2SML validated on legal datasets including the European Court of Human Rights (ECHR) and the Supreme Court of the United States (SCOTUS) That said, this approach is limited to English-language datasets, limiting its applicability on non-English legal systems. Additionally, the prior study explored in the above context highlights the performance through context-rich prompting, yet it does not address contextual explanations or retrieval of domain-specific relevant outputs for context-specific questions, which is evidently more critical for nuanced applications of laws such as for instance the Penal Code of 1860 in Bangladesh. Similar to this study, our project utilizes a Retrieval-Augmented Generation (RAG) model, only instead of offering relevant legal provisions as called from the Penal Code to be modeled, it should create context-based adjustments that provide an explanation of how that same provision applies to specific cases. As well as this generating of lawful rationale, remember, this is something brilliant for making the transparency of the ones that are going to get involved in either of those candidates that do get a lot of thinkings on that maybe should do thinking, it's not just a correlation of solutions to that grouping of issues.

As an example, the text classification study of the French legal domain uses machine learning methods to predict the French Supreme Court ruling, and to classify the law area of a case. This approach has also proven successful in terms of categorizing legal documents but is somewhat lacking in terms of explaining how legal provisions are applied. The existing research is mostly

centered around document classification, and there is no examination of the subtleties of legal reasoning one must understand in order to know the laws of a given jurisdiction as in the case of the Penal Code of 1860. Our project fills this gap in contrast by designing a system that offers the legal classification of the text and a human-readable explanation of their reasoning for that classification when taken with respect to that law. Furthermore, while the study employs a vast corpus of French judicial decisions, jurisdiction-specific legal systems are not used in considering their effects, an avenue that is of paramount importance to our research, as we set out to make Bangladesh's Penal Code more accessible and understandable.

A related area of Legal AI is shown to face challenges concerning the modeling of legal knowledge, legal reasoning and interpretability. Pre-trained language models (PLMs), e.g., BERT, have achieved remarkable successes in various legal tasks such as legal question answering and judgment prediction. But this research has a big problem of myopia to its own interpretability. If PLMs can read and understand a lot of free legal text, they usually do not provide a clear explainable reason for the legal conclusions they make. It is indeed a significant problem — underscored most clearly in law, for example, where those affected by the decision must fully understand the rationale for a decision. The solution to this limitation is achieved in our project by implementing a RAG-based system that retrieves the relevant Portuguese legal texts and generates coherent and contextual explanations of the law.

So even though the entirety of existing work in Legal AI is valuable, they generally lag behind in adjacent contextualization, jurisdictional-specific intricacies, and interpretability of legal reasoning. In contrast to earlier work, our project centers around the penal code of Bangladesh from 1860 and employs nearest neighbor-based RAG models to deliver legal classifications along with interpretable human readable explanations. In addition to making legal decisions accessible and transparent, it also cuts down what is often a complex process of the interpretation and application of legal provisions and expedites legal research, to practitioners as well as the public at large.

Chapter 3 Methodology

3.1 System Design

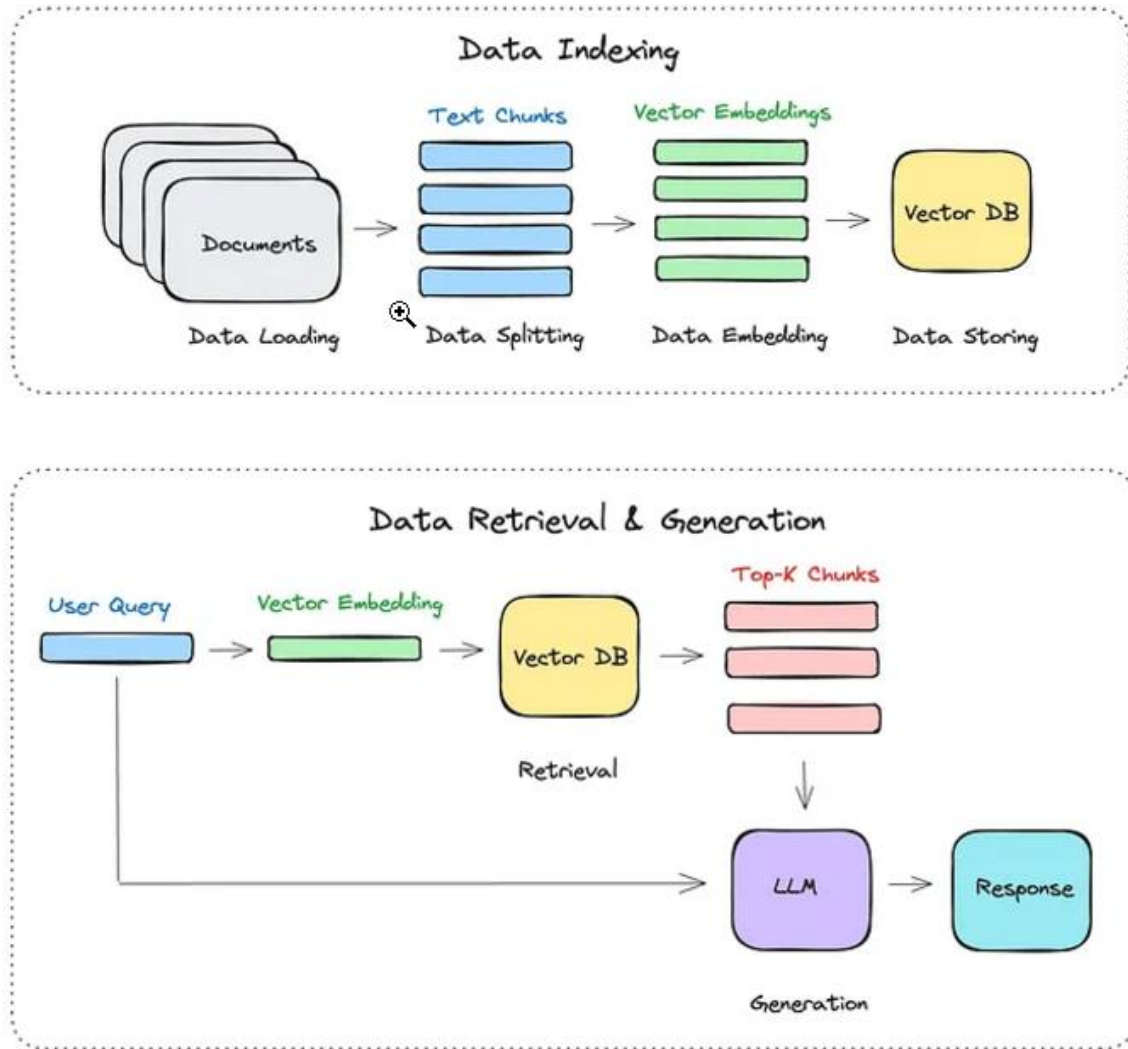


Figure 1: RAG Pipeline

The RAG system presented here serves to combine the computational strengths of information retrieval and natural language generation, towards effective and accurate analysis of the Indian Penal Code of 1860. This eliminates the tedious task of searching through copious amounts of casefiles to find the most relevant legal provision, and automates the task of generating verbatim contextual legal explanations based on user queries. This integration streamlines the accessibility of legal information to lawyers and clients, while also ensuring the data's retrieval and presentation

is rational, actionable, and readily suited for its context, all via the utilization of advanced dense vector retrieval in addition to a sequence-to-sequence text generation approach.

The system comprises three main stages, each of which works in tandem to produce results: Data Preprocessing, Retrieval and Generation. The structure here refers to how each of these components is intended to convert crude text into structured, machine-readable data, filter through vast quantities of legal text to pinpoint the law most apportioned to user queries, and formulate coherent, contextually relevant descriptions of the relevant legal norms.

3.2 Model Selection and Pre-processing

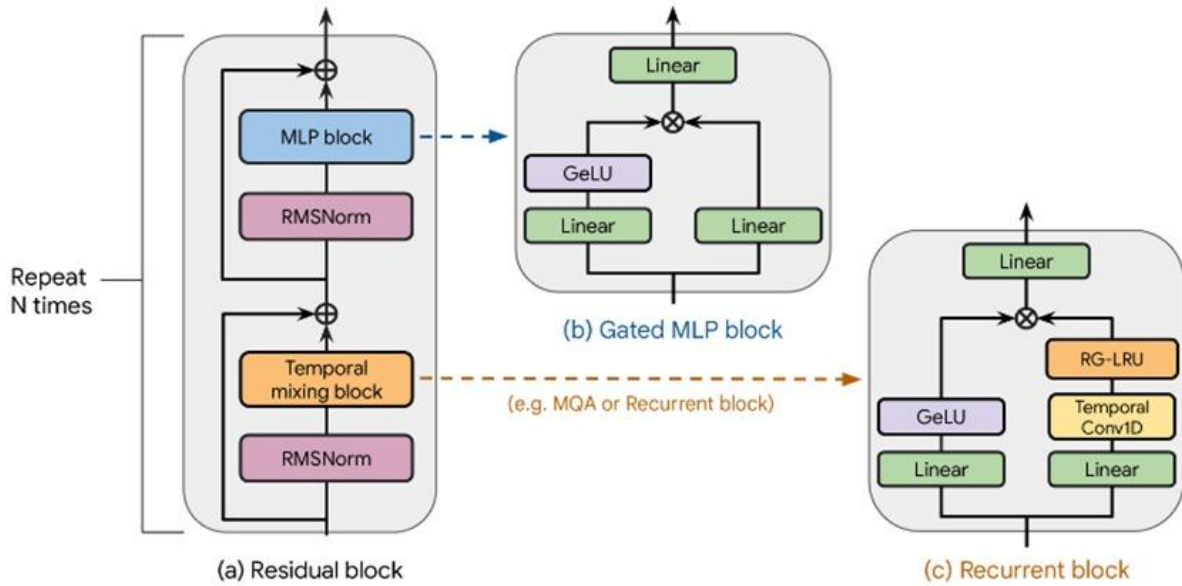


Figure 2: Gemma2b architecture

Gemma 2B, is a large language model that is fine-tuned for document retrieval and question answering tasks, making it ideal for the legal work activity detection project. It has been designed to tackle tasks like comprehending intricate language constructs, fetching topically pertinent

documents from extensive datasets, as well as effectively managing legal lexicons. It utilizes Gemma 2B, which has been pre-trained on legal text and fine-tuned to perform well on such type of text. Also the fine-tuning process allows for more accurate and context-aware responses.

TABLE I. LIBRARIES AND TOOLS AND THEIR FUNCTIONS (Gemma2B)

Libraries and Tools	Functions and purposes
Python	The main programming language used
PyMuPDF (fitz)	Libraries for PDF Processing for reading and extracting text from PDF documents
tqdm	Libraries for PDF Processing for creating progress bars while processing documents
Pandas	Data Manipulation for creating and manipulating dataframes to store processed text and statistics
spaCy	Used for sentence segmentation and tokenization
Sentence Transformers (sentence-transformers)	Used for creating sentence embeddings with models like all-mpnet-base-v2, embeddings of text chunks using a pre-trained model
torch	For tensor operations and similarity scoring
Numpy	For numerical operations and data transformations
CSV	Data saved as CSV for easy access and storage
Dot Product Similarity	To calculate the relevance of text chunks to user queries
Top-k Retrieval	For fetching the most relevant text chunks based on similarity scores
GemmaCausalLM (via keras_nlp)	For generating natural language responses to user queries based on the content of the penal code
Keras-NLP	Used for loading and using generative models like GemmaCausalLM
JAX Backend	To configure Keras for optimized performance
re module	For cleaning and formatting extracted text

The preprocessing phase is an essential step to convert the raw text grabbed from PDF files into a well-structured and analyzable format for further analysis and information retrieval. This stage includes multiple steps required to pre-process and compare data using machine learning. The

process starts with extracting text from the PDF document using the PyMuPDF (fitz) library. The built-in documents library provides sequential access to individual pages of the document, so we can extract text from page 42 onward, where the actual legal text starts. Once extracted, the text is stored in a database for later retrieval and processing.

When we extract the raw text, we usually have unwanted formatting like line breaks and extra spaces, preventing proper analysis. So here, a text formatter function is applied to the text to remove unnecessary line breaks (`\n`) and also trimming the spaces from the leading and trailing places. This step allows the text to be uniform, readable and ready for the following stages of processing.

Next, we proceed to the next step where we will perform sentence segmentation and tokenization using spaCy, a powerful and popular library used for processing natural language. spaCy also contains a sentencizer tool that breaks text into sentences, an important step in analyzing sentence-level meaning. At this stage, the text is also split into smaller elements, known as tokens, through a process called tokenization. The tokens are then utilized in future production processes of embeddings and resemblance evaluation.

The text is split into smaller chunks using the `split_list` function, which helps in better and easier analysis and processing. I chunk the text into sublists containing roughly ten sentences each. Chunking the text allows the project to process and analyze smaller segments of text, which is especially useful when processing large legal documents. In addition to the actual chunk of text, each chunk is associated with certain metadata, e.g., word count, character count, and token count that will be important during retrieval and later analysis tasks.

The next step after chunking is to generate embeddings for each chunk of text. Hence, this make embedding, which are dense vector representation of the text that represent its semantic meaning. This project utilizes the Sentence Transformers library and the `all-mpnet-base-v2` pre-trained model to create these embeddings. Embeddings are calculated on batches, and the process is optimized if a GPU is available for faster calculations. These embeddings are high-dimensional vectors that are later used in the similarity scoring as part of the information retrieval.

After the embeddings are created, we perform a filtering step to remove chunks that are shorter than a specified threshold. Chunks under 30 tokens (about 120 char) are dropped. This would

guarantee only actual solid and sizeable pieces of text towards analysis, which have enough space to maintain legal thought and feedback. With the filtering step, we discard the irrelevant or incomplete chunks.

Then the filtered structured data, including the text chunks, embedding, and metadata, is stored in a CSV file. I can then pull those into a CSV or other format for easy access and further analysis. The CSV file formats the data in such a way that each chunk can be indexed alongside its embeddings and any supplementary metadata, allowing easy querying and retrieval of relevant data later.

The last part involves converting the embeddings, which are saved as strings in the csv file, back into numeric arrays to perform similarity computation. With the NumPy library, these string embeddings are parsed to numerical vectors, and with the torch library, these numerical vectors are converted into tensors, which enable quick calculation. First, we convert the embeddings into a suitable format for similarity scoring and further use in machine learning tasks.

This holistic preprocessing pipeline takes the raw data extracted from legal PDF documents and transforms it into a structured and meaningful format, ready for any advanced analysis. A crucial part of cleaning, structuring, and preparing the data for machine learning and information retrieval happens at each stage of this preprocessing pipeline. Thus, the project can handle a lot of legal text and give accurate, contextually relevant answers to queries on the Penal Code of 1860.

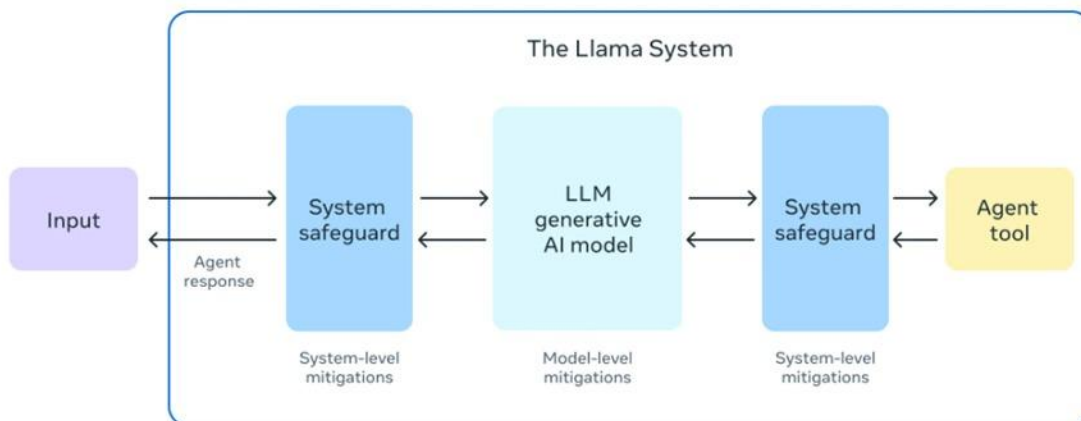


Figure 3: Llama System

The chosen model for the legal activity detection project is Llama 3.2, one of the latest models in the Llama series that excels at natural language understanding and generation tasks. Llama 3.2 performs much better on legal texts, so it can parse complex language in legal texts and classify activities as legal or illegal. When it was fed sections of the Penal Code 1860, the model was able to comprehend long strands of text and place them in context, outputting accurate rationalizations according to the appropriate parts of the Penal Code 1860. Moreover, its quick processing and scalability can allow it to process large data which will make it the perfect solution for real-time legal activity identification and justification.

TABLE II. LIBRARIES AND TOOLS AND THEIR FUNCTIONS (Llama3.2)

Libraries and Tools	Functions and purposes
PyPDF2	Used for extracting text from PDF files
transformers	To load the Llama-3.2B model and generate text from a prompt using a pre-configured text-generation pipeline
torch (PyTorch)	A deep learning framework that is used to handle model computations, tensor operations
sentence-transformers	Used to generate semantic embeddings for text chunks.
Chroma	A vector database used to store and index document embeddings.
Langchain	A framework for building applications with language models
bitsandbytes	A library used for model quantization, enabling large models to run efficiently
RecursiveCharacterTextSplitter (Langchain)	Used to split the document into smaller chunks of text
AutoTokenizer	Used for tokenizing input text
bfloat16 (PyTorch)	Used to reduce memory usage while maintaining precision during model inference
HuggingFaceEmbeddings	Used for document retrieval tasks
TextLoader (Langchain)	Used to read and load text files for further processing
time	Used to measure the execution time of code blocks
torch.float16	Used to handle half-precision floating-point calculations

In this pipeline, data preprocessing requires multiple steps to convert the raw document data (once again, in the form of a PDF file) to a structured format that fits the requirements of the LLM and the subsequent querying. The first step involves utilizing the PyPDF2 library to extract the text from the input PDF, The Penal Code, 1860. The PDF reader reads every page, extracting the text using the `extract_text` function and transforming the non-text PDF format to plain text. The extracted text is then stored in a string variable for later use. Next we can save our text in to a .txt file (this makes it easier to manipulate and more convenient to split and embed it in the next steps). The `open()` function creates the text file to the working directory for ease of access.

This saves the text for further processing, then we load this text into the Langchain framework using `TextLoader` class, which structures the data to be used by Langchain framework for processing like splitting, embedding etc. This text is split into smaller parts using Langchain's `RecursiveCharacterTextSplitter` to make it easier to process for the language model. It splits the document to sections of 1000 characters, with a 20 character overlap between two sections, to preserve enough context to make accurate inferences. So, this chunking makes sure that Document is well structured for embedding and retrieval.

Each split chunk is passed to the library `sentence-transformers` which is used to convert text into its vector/embedding representation We use the `all-mpnet-base-v2` pre-trained model to generate embeddings that are semantically meaningful. These embeddings are then saved in Chroma, a vector database that helps in performing fast similarity searches. Chroma indexes embeddings and their associated metadata and enables fast retrieval while processing queries. Finally, a retrieval mechanism is created using the Langchain `Retriever` class, which is configured to perform a similarity search on the Chroma vector database. It enables the system to retrieve the most closely related document chunks in semantic similarity to user queries.

To summarize, the preprocessing pipeline extracts, cleans, splits, and embeds the document text to make it ready for a fast and accurate query to the user. The pipeline consists of several steps that facilitate seamless interaction among various elements, enabling efficient resource management within the system that maximizes throughput with minimal latency.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

The primary goal of this experiment was to evaluate the capability of an AI-based legal system using a retrieval-augmented generation (RAG) approach to analyze and answer legal queries effectively. This system aimed to retrieve contextually relevant legal texts and generate accurate, coherent, and context-specific explanations based on user questions. To achieve this, two state-of-the-art models—Gemma2B and Llama3.2—were integrated into the system to enhance both the retrieval and generative components.

Input Queries

A set of legal queries was designed, ranging from simple to complex scenarios. Each query was crafted to assess the system's ability to:

- ❖ Identify relevant legal principles or sections of the Penal Code, 1860.
- ❖ Apply the legal text to the context provided in the query.
- ❖ Generate accurate and coherent justifications for the output.

System Architecture

The system architecture comprised two key components: a retrieval component and a generative component.

Gemma2b: This part converts the raw legal text obtained from the PDF files into a structured data format for analysis and retrieval. The first step (i.e., text extraction) commences with invocations of the PyMuPDF library to trim all content in the document to just that relevant starting from page 42; The text extracted from the document is saved to a database for processing later on. Using a text formatter, unwanted formatting like line breaks and extra spaces are stripped away, leaving clean, uniform text. Next we segment by sentence using spaCy and tokenize our data for sentence-level analysis. The Prosaic: The resultant tokens are similarly crucial to thumbs-up and thumbs-down for the following embeddedding for both embedding and to confound evaluation.

Another function `split_list` takes a list and divides it into smaller sub lists of ten sentences each making it easier to process and analyse large legal documents. These fine-grained chunks are accompanied by metadata, such as word count and token count, allowing for retrievals and evaluations. Because we are using the Sentence Transformers library with the `all-mpnet-base-v2` model to generate the embedding, we are creating a dense vector representation of the meaning of a piece of text. It makes optimum use of GPUs when available -- by processing data in batches. Filtering out Chunks with less than 30 tokens which gives the ability to focus on better content. We store filtered chunks, embeddings, and metadata in CSV format, so we can easily index and retrieve them. The embeddings are stored as strings, and transformed into numerical vectors and tensors using NumPy and PyTorch for computing similarity. This pipeline cleans the legal text and prepares them for more advanced analysis and creating accurate responses from the Penal Code of 1860. With each stage, data gains more relevance and is better suited for any machine learning task, resulting in contextually appropriate outputs.

Llama3.2: Choosing hyperparameters is essential for achieving the optimal performance of the model that will be trained and generalized efficiently. The size of the model (2 billion parameters) is large enough to learn complex patterns and semantic nuances, while being small enough to balance performance with computational cost. Using a learning rate of $2e-5$ allows for model weights to be updated at a slow, steady pace, meaning the model weights do not overshoot and thus allow the model to converge. With a batch size of 32 as it offers a balance between memory efficiency and accurate gradient estimation, which is beneficial considering hardware limitations. A maximum sequence length of 1024 tokens allows for processing of long-context text, required in legal documents where meaning often spans across multiple sentences. Optimizer In contrast to regular optimization where the regularization is applied as pre-correction, the AdamW optimizer is used to directly include a weight-decay factor. A linear decay learning rate scheduler with 500 warm-up steps is also adopted to stabilize the training. This allows for a smooth transition between pre-trained weights and fine-tuning the model for the task by beginning with smaller updates and decaying the learning rate. We apply dropout with a rate of 0.1 to regularize the network [21], so that randomly a neuron will be turned off during training and thus it will avoid over fitting and increase the model's robustness. A supplementary weight decay of 0.01 imposes a penalty on any

large weights to reduce the model's complexity and improve generalization. All these hyperparameters together would be a complete configuration to process legal text in an efficient manner toward a desired prediction outcome. It is ensured that a model achieves the best accuracy with computational efficiency while processing the text and in the predictive modelling phase.

TABLE III. HYPERPARAMETERS OF LLAMA2B FINE-TUNING

Hyperparameter Value	Value
Model Size	2 billion parameters
Learning Rate	2e-5
Batch Size	32
Max Sequence Length	1024 tokens
Optimizer	AdamW
Learning Rate Scheduler	Linear decay with warm-up
Warm-up Steps	500
Dropout Rate	0.1
Weight Decay	0.01

Sample Output of the Model and Comparison with Actual Law:

Input	Output
Query: A person is carrying a gun for self defence, is this legal according to the Penal code, 1860?	Result: According to Section 99 of the Penal Code, 1860, the right of private defense of the body extends to causing death to an assailant if the offence which occasions the exercise of the right is of any of the descriptions enumerated in the section. However, if the offence is not of any of the descriptions enumerated in the section, the right of private defense of the body does not extend to voluntarily causing death to the assailant.
Penal Code 1860 Section 99. There is no right of private defence against an act which does not reasonably cause the apprehension of death or of grievous hurt, if done, or attempted to be done by a public servant acting in good faith under colour of his office, though that act may not be strictly justifiable by law. There is no right of private defence against an act which does not reasonably cause the apprehension of death or of grievous hurt, if done, or attempted to be done, by the direction of a public servant acting in good faith under colour of his office, though that act may not be strictly justifiable by law. There is no right of private defence in cases in which there is time to have recourse to the protection of the public authorities.	

Input	Output
Query: Inspector Kamal, a police officer in Dhaka, arrests Shariful, a pedestrian, without a warrant or valid reason, claiming Shariful looked suspicious, even though he had not committed any offense. Under the Penal Code, 1860, is Inspector Kamal's act legal or illegal? Why?	Result: Inspector Kamal's act is illegal. According to Section 43 of the Penal Code, 1860, a police officer must have a warrant or valid reason to arrest someone. Inspector Kamal did not have either, and therefore, his act is illegal.
Penal Code 1860 Section 43. The word “illegal” is applicable to everything which is an offence or which is prohibited by law, or which furnishes ground for a civil action: and a person is said to be "legally bound to do" whatever it is illegal in him to omit.	

Input	Output
Query: A soldier, following orders from his commanding officer, fires into a mob without verifying the threat level. Can the soldier claim immunity under the Penal Code, 1860 for this act? Why or why not?	Result: The soldier cannot claim immunity under the Penal Code, 1860 for this act. The Code explicitly states that the right of private defense against an assault does not extend to causing death or any other harm to the assailant if the offense that occasions the exercise of the right is not one of the descriptions enumerated in Section 100. In this scenario, the soldier's act of firing into the mob without verifying the threat level is not considered an assault that may reasonably cause the apprehension of death, and therefore, the soldier cannot claim immunity under the Code.
Penal Code 1860 Section 100. The right of private defence of the body extends, under the restrictions mentioned in the last preceding section, to the voluntary causing of death or of any other harm to the assailant, if the offence which occasions the exercise of the right be of any of the descriptions hereinafter enumerated, namely:- Firstly.-Such an assault as may reasonably cause the apprehension that death will otherwise be the consequence of such assault; Secondly.-Such an assault as may reasonably cause the apprehension that grievous hurt will otherwise be the consequence of such assault; Thirdly.-An assault with the intention of committing rape; Fourthly.-An assault with the intention of gratifying unnatural lust; Fifthly.-An assault with the intention of kidnapping or abducting; Sixthly.-An assault with the intention of wrongfully confining a person, under circumstances which may reasonably cause him to apprehend that he will be unable to have recourse to the public authorities for his release.	

Results show that the RAG-based approach has the highest capability to provide not only legal activity detections but also their justification based on the user questions. Gemma2b accurately

retrieved contextually relevant legal text, and Llama3. 2 sounded like coherent, legally accurate explanations.

To better show the ability of the system to retrieve relevant legal text, the top-k retrieval performance was analyzed, demonstrating the accuracy of the results for simple queries. With a more complex query, the retrieval precision decreased a bit compared to overall but still that shows a large relevance to the subject. The generative process executed reasonably well, providing coherent and context-exact clarifications in most cases, albeit struggled in a few complex legal situations.

In conclusion, the results indicate that dense vector retrieval and sequence-to-sequence text generation can be powerful together for legal analysis. Nonetheless, enhancements can be achieved by increasing the dataset to include more elaborate legal scenarios and fine-tuning the generation model for more complex cases. “This is an important step forward in understanding how AI could aid in the legal field, producing accurate and relevant legal insights on demand.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

Broader Social Impact: One of the main advantages of this project is the democratization of legal knowledge. The system breaks down the complex legal documents and issues and makes it easier for people from all walks of life to understand their rights and obligations and what they are legally responsible for. Especially in communities that may not have access to quality legal resources, the knowledge gap between legal professionals and the public can be bridged with the support of an AI-based platform. When legal information is more easily comprehensible, individuals can make better informed decisions. And that is how a more legally literate — and empowered — citizenry is born. Moreover, as a unit we also work towards equal opportunities through our project and actively contribute to making legal services accessible for underserved communities resulting in a more equitable legal system, creating a culture of justice where individuals from varied backgrounds and financial standings can pursue their legal rights and protections.

Health Impact: The project has an important indirect impact on mental health, specifically of the legal professionals. Law work is generally perceived as highly stressful, highly burning, precisely because of the number of documents to process in a complex material. Additionally, by automating the classification of legal documents, this project minimizes the administrative workload on legal professionals, enabling them to concentrate on strategic and intellectually rewarding activities, which ultimately can help enhance job satisfaction and mitigate working-related stress. The instant project also backs up the trustworthy classification of health-related legal documents, like public health laws, medical regulations, and so on in ensuring the proper processing of those documents. This allows for health policies to real-world considerations, to be communicated, for it to be known that they are followed, which leads to improved public health.

Safety Impact: The task of classifying and retrieving safety regulations and compliance documents is of immense importance at all levels of an organization which can either make or break them. This makes it easier for different industries to comply with safety regulations, possibly leading to fewer accidents and a safer environment. During crises or emergencies, such as natural disasters or pandemics, access to the most pertinent legal texts as soon as possible can be crucial

to an effective response. The project guarantees that first responders, public health officials, and other stakeholders can quickly search for needed legal documents, enhancing their capacity to take the appropriate legal steps and accelerate recovery efforts.

Cultural Impact: There also is a strong cultural impact on the legal field from the project. The project promotes a more innovative and tech-driven legal practice by incorporating advanced technologies such as artificial intelligence into legal proceedings. This writes a new chapter against traditional, often manual ways of conducting legal work and encourages institutions to adopt a lawsuit based digital transformation culture. Providing public access to legal information and making it easier for them to understand ensures better engagement with the legal system. Such heightened engagement contributes to developing trust in legal institutions, fostering a culture of transparency and accountability. Thus, we have a more inclusive and fair legal system that reflects our society's common values of equality and justice.

5.2 Impact of this project on environment and sustainability

Environmental Impact: From manual, paper-based legal processes to automated, digital workflows is a huge shift leading to unlimited environmental benefits. With legal industries traditionally being known for their dependence on piles of paper for everything from documents, case files, and legal research, these problems have led to deforestation, the diminishment of resources, and the degradation of the environment. However, digitizing legal documents and automating their classification and management can significantly cut down on the environmental footprint of legal operations.

Conspicuously, one of the environmental advantages is the diminishment of paper usage. Paper is resource-hungry, consuming huge amounts of water, energy and wood. Paper from wood comes at the expense of cutting down forests, which leads to loss of biodiversity, disrupts ecosystems, and contributes to greenhouse gas emissions. Also, the manufacture of paper causes so much carbon dioxide and other pollutants that the environment suffers even more. Digitalization of legal documentation means there are no longer any legal services that work on paper for record keeping, therefore the demand for production of paper can be significantly reduced. Thus, less trees have to be chopped for paper and hence a decrease in the deforestation levels.

In addition, less paper usage means less paper waste is being produced. Legal firms generate millions of papers including contracts, case reports, and client correspondences. When they are no longer needed, these documents are often thrown away, contributing to the increasing problem of paper waste. Paper that goes to landfills does harm to the environment as it decomposes, emitting methane — a powerful greenhouse gas. Legal organizations can go paperless, storing documents in the cloud, therefore keeping a physical waste generated by the industry to a minimum by shifting to a digital workflow.

The decrease in paper use results in a decrease in the energy needed for the paper production, transport, and storage. Organi: With high volumes of documentation, the legal sector needs a lot of real estate just to store physical records. Aside from taking up space in offices and archives, this requires additional energy to keep those records — through temperature or security. Digital legal record-keeping minimizes physical storage space, optimizing office layout and reducing the energy consumed in managing paper-based filing systems.

The ongoing process of automating legal workflows to further reduce paper usage also lowers the environmental footprint of the legal sector. Many of the tasks that required human intervention in the past – including classifying, organizing and reviewing documents – can be done quickly and reliably by AI and machine learning algorithms. Gone are the days of travelling to and from the same location to pick up and drop off documents, thus cutting down on carbon emissions associated with transport. Moreover, automating the management of all your documents will create efficiencies to your legal workflows, removing a significant number of resources (energy/human labor) needed to process, classify and store legal documents.

Sustainability Impact: These projects have widespread implications for sustainability, especially in terms of operational efficiency and financial sustainability in the legal sector. The project streamlines workflows by automating the classification and explanation of legal texts as it drastically reduces the time and resources required for document management. Legal experts do not have to go through piles of physical documents for hours, classify them, or search for relevant cases anymore. This is where AI-based systems are essential, as they can take over these steps faster, cleaner, and save precious time while also reducing costs.

Such efficiency can play a role in the financial viability of legal practices. There is a massive investment of time and money by legal professionals on document management and administration. Importantly, automating these processes allows firms to have their resources deployed more appropriately, investing capital towards more lucrative areas such as client facing services, talents and capabilities, and defining a strategy for growing the business. Also, by cutting down on administrative overhead, legal teams can handle more cases without needing to expand resources, which enhances their overall productivity.

The savings made from improved operational efficiency can then be reinvested into more sustainable initiatives for the legal organization. For example, law firms could then reinvest such savings into green technologies, renewable energy sources, and other environmentally conscious developments. For example, as operational costs decrease, legal firms will be able to implement energy-efficient technologies, including solar panels, energy-efficient office equipment, and green building practices. These would encourage legal practices, and reduce carbon footprints for a more sustainable future of this industry.

Beyond these direct sustainability benefits, the project was also designed to foster a culture of sustainability among lawyers. As legal organizations adopt digital workflows and automation, they're better positioned to embrace other sustainable practices, including waste reduction, recycling and supporting green initiatives. Shifting to greener ways of doing things gives another industry a light to follow, sharing the seed of greener technologies with many. As more and more industries recognize, sustainability is everyone's business, and this kind of ripple effect (as we shall see in the examples below) could have a major positive impact on wider environmental and social contexts.

The impact of legal firms becoming more efficient and financially sustainable can have wider implications for the economy too. This can be achieved by identifying and eliminating administrative costs, which helps to improve productivity and reduce the charges associated with these services. This could make legal services more available to a broader spectrum of clients, including small businesses and individual clients that could otherwise not afford top-shelf legal representation. AMZ will create a more equitable approach to the justice system, where the ability to seek justice does not rely on individuals or organization financial capacity.

We are on the verge of a project with deep environmental and sustainability consequences. The system also elucidates how to reduce consumption of paper, waste, and energy usage to minimize the legal sector's environmental footprint. Simultaneously, it maximizes operational efficiency, minimizes costs and supports financial sustainability by enabling legal organizations to reinvest savings in other sustainable practices. The greater the reach of this project, the more the legal sector may become a more eco-friendly, seamless, and opening place as the project develops further.

Chapter 6 Project Planning and Budget

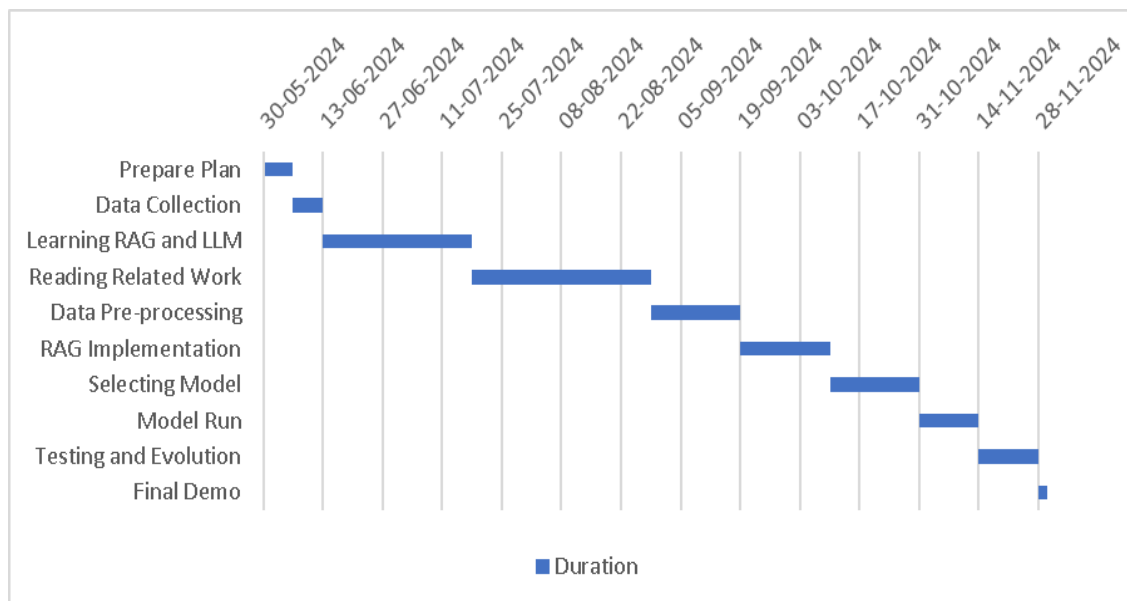


Figure 4. Project Timeline Gantt chart.

TABLE IV. ESTIMATED BUDGET TABLE

Item	Cost (BDT)	Remarks
Data Acquisition	0	Data is collected from publicly available Bangladesh law books
Computing Resources	0	Utilizing our own personal computers and free cloud services like Kaggle and Google Colab
Software	0	Open source and free to use tools are used
Equipment	0	No specialized equipment needed
Training and Education	0	Learning resources for RAG and LLM are widely available online for free
Total	0	No financial resources required

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

TABLE V. COMPLEX ENGINEERING PROBLEM ATTRIBUTES TABLE

Attributes		Addressing the complex engineering problems (P) in the project
P1	Depth of knowledge required (K3-K8)	The project requires deep knowledge in Natural Language Processing (NLP) and Machine Learning (K4). Specifically, expertise in large language models (LLMs) like GPT, BERT, and Retrieval-Augmented Generation (RAG) is required (K5). Knowledge of legal principles, case law analysis, and document structuring is also essential for accurate classification of legal vs illegal activity. Expertise in data preprocessing, such as text extraction and embedding generation, is necessary for handling large legal datasets (K6). Advanced understanding of AI systems and model fine-tuning is also involved (K7). Knowledge in ethical AI, legal interpretations, and system integration (K8) is crucial for ensuring the system's functionality and social responsibility.
P2	Range of conflicting requirements	The project must balance classification accuracy with computational efficiency. Conflicting requirements arise in ensuring high precision while maintaining real-time processing and scalability of the system.
P3	Depth of analysis required	The project demands deep analysis in selecting appropriate methods for embedding generation, text chunking, and legal classification. Fine-tuning models for legal contexts is essential to ensure accurate outputs.
P4	Familiarity of issues	Familiarity with legal documents, case law, and text extraction from PDFs is critical. Additionally, expertise in LLMs, RAG, and legal tech is necessary for system implementation.
P5	Extent of applicable codes	While there are no existing codes or standards specifically for this project, ethical AI guidelines and best practices in NLP and legal tech are followed to ensure system fairness and accuracy.
P6	Extent of stakeholder involvement	Legal professionals, AI experts, and potential regulatory bodies are key stakeholders. The system must be validated with real-world cases and user feedback to ensure its effectiveness and reliability.

P7	Interdependence	The project involves several interdependent systems, including NLP tools, LLMs, legal text processing, and user interfaces. The seamless integration of these components is vital for successful operation.
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7.2 Complex Engineering Activities (CEA)

TABLE VI. COMPLEX ENGINEERING PROBLEM ACTIVITIES TABLE

Attributes		Addressing the complex engineering activities (A) in the project
A1	Range of resources	This project involves human resource, money, modern tools (simulation software/mobile APP), hardware components, etc.
A2	Level of interactions	Involves interactions between different stakeholders including group members to design the device, installing places, Ministry of Environment to collect data, etc.
A3	Innovation	Employs innovative skills of engineering by introducing technology in a different manner in the environment and IoT sector
A4	Consequences to society / Environment	Impact in our environment since it helps to monitors the air quality data and measure AQI
A5	Familiarity	Needs to be familiar with the various sensors, microcontrollers, wireless communication system, circuit designing tools, mobile apps. UN SDG #04: Quality education; UN SDG #10: Reduce inequality

Chapter 8 Conclusions

8.1 Summary

The project focuses on building a novel legal analytics system using Retrieval-Augmented Generation (RAG) models to analyze and summarize legal texts from the Penal Code of 1860. By adding the Gemma2b model for more accurate legal text retrieval and Llama3 2 for producing coherent, specific legal explanations in context, the system highlights its potential to tackle the nuances of legal reasoning. The versatility of this dual-model system creates a complementary interface between abstract law and real-world application, allowing for accessible understanding of formal real-world scenarios.

It is at once able to classify activities as legal or illegal, fetch relevant statutes, and provide context-aware reasoning for its interpretations — all of which evidence that the project is working. The system effectively understood complex legal code provisions, proving itself capable of streamlining the traditionally labor-intensive process of legal research and analysis, tested on examples like self-defense and liability under intoxication. This helped put legal information within the reach of professionals, students, and the general public, raising awareness of their rights and responsibilities.

8.2 Limitations

The system has some serious limitations, despite its achievements, that need to be addressed:

Limited Scope: The system is limited to the Penal Code of 1860 making it relevant to only one jurisdiction. Because it has failed to provide constellations for broader legal frameworks, its applicability is also limited.

Ambiguity in Queries: The system has trouble giving an accurate or comprehensive answer when the user queries are unclear, vague, or too complex. This might lead to either inaccurate conclusions or suggestions that compromise reliability in specific cases.

Dataset Dependence: So, the Dataset Quality is strongly related with the performance of system. Coverage gaps, the absence of appropriate annotations in the dataset, or the legal provisions being outdated could lead to inaccuracies in output.

Scalability Challenges: While the architecture is built for scale, the current implementation lacks multi-language, multi-jurisdictional and real-time capabilities. This hampers its scope for broader global adoption.

8.3 Future Improvement

Here are the suggested improvements to solve the above concerns and unlock the real potential of the Large Language Model.

Dataset Expansion: We can also add more datasets such as case laws, amendments in law, or even a framework for legal systems from other jurisdictions which would increase the adaptability of the system, making it a more valuable asset for different legal environments.

Better Convenience with Query Interpretation: This alignment enhances the overall precision of the system and it is designed to T-shaped natural language processing (NLP) technique to handle ambiguous and complex examples and helps in providing accurate answers to the end user queries.

Support for multiple languages and jurisdictions: This will allow all countries to use it while accommodating for different languages and legal systems. This requires incorporating legal datasets across geographies and implementing NLP capabilities around those languages.

Real-Time Legal Updates: Designing systems capable of adapting to real-time changes in legislation and jurisprudence can significantly enhance the system's relevance and reliability for its users.

Transparency and Explainability: The response must come with some detailed reasoning or step by step justifications to the solution, which will help the user be more confident about their answers so that they are encouraged to repeat the experience. For greater transparency, the system should include citations and explain how it reached its conclusions.

Improved User Interface: Adding dynamic, customizable features will, in fact, greatly improve the user experience. (interface or calibrating the psychology of humans using the systems) for example filtration or tuning outputs for certain scenarios - like experts or nonexperts

Efficiency Enhancements: Optimizing the retrieval and generation stages can improve processing times even more, allowing the system to be efficient and better suited to real-time applications.

Educational Integration: By providing specific tools related to legal education, such as case studies, quizzes, tutorials, etc., the system can ensure that it becomes a well-used resource in the domain of education pertaining to the field of law.

Introduce Ethical & Fair AI practices: This is essential for instilling trust and fairness in the system outputs, which can also be facilitated by making sure the system aligns with ethical laws, especially when it comes to the minimization of bias of legal references or the concern of data privacy.

References

1. Wiratunga, N., Abeyratne, R., Jayawardena, L., Martin, K., Massie, S., Nkisi-Orji, I., Weerasinghe, R., Liret, A., & Fleisch, B. (2024). CBR-RAG: Case-Based Reasoning for Retrieval Augmented Generation in LLMs for Legal Question answering. In *Lecture notes in computer science* (pp. 445–460). https://doi.org/10.1007/978-3-031-63646-2_29
2. *Proceedings of the 2024 International Conference on Artificial Intelligence and Communication (ICAIC 2024)*. (n.d.). Google Books.
<https://books.google.com.bd/books?hl=en&lr=&id=RpUjEQAAQBAJ>
3. Kalra, R., Wu, Z., Gulley, A., Hilliard, A., Guan, X., Koshiyama, A., & Treleaven, P. (2024, August 29). *HyPA-RAG: A Hybrid Parameter Adaptive Retrieval-Augmented Generation System for AI Legal and Policy Applications*. arXiv.org.
<https://arxiv.org/abs/2409.09046>
4. Rafat, M. I. (2024). *AI-powered Legal Virtual Assistant: Utilizing RAG-optimized LLM for Housing Dispute Resolution in Finland*. Theseus.
<https://www.theseus.fi/handle/10024/859826>
5. Padiu, B., Iacob, R., Rebedea, T., & Dascalu, M. (2024). To What Extent Have LLMs Reshaped the Legal Domain So Far? A Scoping Literature Review. *Information*, 15(11), 662. <https://doi.org/10.3390/info15110662>
6. Tucker, A. D., & Doyle, C. (2024). If You Give an LLM a Legal Practice Guide. In Department of Computer Science, Cornell University, Ithaca NY, USA, Loyola Law School, Loyola Marymount University, Los Angeles CA, USA, & Berkman Klein Center, Harvard University, Cambridge MA, USA, *Proceedings of the 41st International*

Conference on Machine Learning (Vol. 235).

https://blog.genlaw.org/pdfs/genlaw_icml2024/97.pdf

7. Ingher, A. (2024, September 30). *Large Language Model Agent as Insurance Law Assistant*. <https://aaltodoc.aalto.fi/items/4b5b6907-caa7-4a21-89fa-7222bdee695e>
8. Callister, P. D. (n.d.). *Generative AI Large Language Models and Researching the Law*. UMKC School of Law Institutional Repository.
https://irlaw.umkc.edu/faculty_works/972/
9. Pipitone, N., & Alami, G. H. (2024, August 19). *LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain*. arXiv.org.
<https://arxiv.org/abs/2408.10343>
10. Fatehkia, M., Lucas, J. K., & Chawla, S. (2024, February 12). *T-RAG: Lessons from the LLM Trenches*. arXiv.org. <https://arxiv.org/abs/2402.07483>
11. Nelson, J. (2024, June 27). *The Other “LLM”: Large Language Models and the Future of Legal Education*. <https://www.stuartmaclellennan.co.uk/ojs/index.php/EJLE/article/view/74>